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Inventor(s)	Shi; Minglin et al.

Torque wrench with strain gauges

Abstract

An example torque wrench is provided. The example torque wrench may include a drive head configured to engage with a tool for rotating a fastener. The drive head may have a drive axis about which the drive head rotates when rotating the fastener. The torque wrench may also include a deflection member coupled to the drive head and an outer body coupled to the deflection member at a first loading point and a second loading point. The torque wrench may also include a first strain gauge coupled to the deflection member between the drive head and the first loading point, and a second strain gauge coupled to the deflection member between the first loading point and the second loading point.

Inventors:	**Shi; Minglin (Shanghai, CN), Luo; Henglian (Shanghai, CN), Yang; Cheng (Shanghai, CN)**
Applicant:	**Apex Brands, Inc.** (Apex, NC)
Family ID:	**77890788**
Assignee:	**APEX BRANDS, INC. (Apex, NC)**
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Primary Examiner: Thomas; David B.

Attorney, Agent or Firm: BURR & FORMAN

Background/Summary

TECHNICAL FIELD
(1) Example embodiments generally relate to wrenches, and in particular to electronic torque wrench technology.
BACKGROUND
(2) Wrenches are often employed to engage with various types of fasteners to provide a user with leverage, via a handle, to turn the fastener. In some applications, fasteners must be tightened to particular, specified torque. To ensure proper tightening, a torque wrench may be used, which is a wrench that indicates, either mechanically or electrically, that a desired torque has been applied to the fastener. A torque wrench may be set to a desired torque, and the wrench may indicate when that torque setting has been reached when tightening a fastener.
(3) Many electronic torque wrenches use a strain gauge to measure the torque being applied to a fastener by the wrench. However, in conventional torque wrenches, the location where the force on the handle is applied (e.g., close to the head of the wrench or close to the end of the handle) may have an impact on the reading provided by the strain gauge when the same torque is actually being applied. As such, the measurement accuracy of a conventional torque wrench may be limited when the force on the handle is moved away from an ideal force applied location on the handle. Further, in some applications, placement of the force at the exact location on the handle may be difficult or impossible, leading to incorrect torque measurements. As such, there is a need for a torque wrench that is capable of measuring the torque more accurately when the position of the applied force on the handle is moved to different locations.
BRIEF SUMMARY OF SOME EXAMPLES
(4) According to some example embodiments, an example torque wrench is provided. The torque wrench may comprise a drive head configured to engage with a tool for rotating a fastener. The

drive head may have a drive axis about which the drive head rotates when rotating the fastener. Further, the torque wrench may comprise a deflection member coupled to the drive head, and an outer body coupled to the deflection member at a first loading point and a second loading point. The torque wrench may also comprise a first strain gauge coupled to the deflection member between the drive head and the first loading point, and a second strain gauge coupled to the deflection member between the first loading point and the second loading point.

(5) According to some example embodiments, another example torque wrench is provided. The torque wrench may comprise a drive head configured to engage with a tool for rotating a fastener. The drive head may have a drive axis about which the drive head rotates when rotating the fastener. The torque wrench may also comprise a deflection member coupled to the drive head and a handle coupled to the deflection member at a first loading point and a second loading point. The torque wrench may further comprise a first strain gauge coupled to the deflection member between the drive head and the first loading point, and a second strain gauge coupled to the deflection member between the first loading point and the second loading point. Additionally, the torque wrench may comprise processing circuitry electrically coupled to the first strain gauge and the second strain gauge and configured to measure a voltage between an output of the first strain gauge and an output of the second strain gauge. The voltage may be based on a torque being applied to the fastener.

(6) According to some example embodiments, an example method for measuring a torque applied by a drive head of a torque wrench to a fastener is also provided. The torque wrench may comprise a deflection member coupled to the drive head. The method may comprise measuring, by processing circuitry, a voltage between an output of a first strain gauge and an output of a second strain gauge. In this regard, the voltage may be based on a torque being applied to the fastener. The first strain gauge may be coupled to the deflection member between the drive head and a first loading point, and the second strain gauge may be coupled to the deflection member between the first loading point and a second loading point. The first loading point and the second loading point may be points of mechanical coupling between the deflection member and a handle of the torque wrench. The method may also comprise converting the measured voltage into a torque measurement.

(7) According to some example embodiments, a torque wrench may comprise a drive head configured to engage with a tool for rotating a fastener. The drive head may have a drive axis about which the drive head rotates when rotating the fastener. The torque wrench may also comprise a deflection member coupled to the drive head, a handle coupled to the deflection member, and a strain gauge assembly coupled to the deflection member. The strain gauge assembly may be configured to measure strain on the deflection member as an indication of a torque being applied to the fastener by the torque wrench. The torque wrench may further comprise a handle extender coupled to the handle. The handle extender may be configured to be removable from the handle by a user or installed on the handle by the user. The handle extender may be configured to increase a handle length of the torque wrench relative to the handle length without the handle extender coupled to the handle. Additionally, the handle may be coupled to the deflection member at a first loading point and a second loading point. Further, the strain gauge assembly may comprise a first strain gauge coupled to the deflection member between the drive head and the first loading point, and a second strain gauge coupled to the deflection member between the first loading point and the second loading point.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

(1) Having thus described some example embodiments in general terms, reference will now be

made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:
(2) FIG. **1** illustrates an example torque wrench according to some example embodiments;
(3) FIG. **2** illustrates an exploded view of a forward portion of the torque wrench of FIG. **1** providing a view of the deflection member according to some example embodiments;
(4) FIG. **3** illustrates a cross-section view of the torque wrench of FIG. **1** in a fully assembled configuration according to some example embodiments;
(5) FIG. **4** illustrates a cross-section view of the torque wrench of FIG. **1** being subjected to an applied force according to some example embodiments;
(6) FIG. **5** illustrates a force diagram for the drive head and the deflection member resulting from the force applied in FIG. **4** according to some example embodiments;
(7) FIG. **6** illustrates a cross-section view of a torque wrench with a handle extender according to some example embodiments;
(8) FIG. **7** illustrates a schematic of an electrical configuration of strain sensors according to some example embodiments;
(9) FIG. **8** illustrates a block diagram of an electronics assembly for the torque wrench of FIG. **1** according to some example embodiments;
(10) FIG. **9** illustrates a flowchart of an example method for measuring a torque applied by a torque wrench to a fastener according to some example embodiments;
(11) FIG. **10** illustrates a graph of torque measurements taken with a force applied at different locations on a conventional torque wrench; and
(12) FIG. **11** illustrates a graph of torque measurements taken with a force applied at different locations on a torque wrench according to some example embodiments.

DETAILED DESCRIPTION

(13) Some example embodiments now will be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all example embodiments are shown. Indeed, the examples described and pictured herein should not be construed as being limiting as to the scope, applicability, or configuration of the present disclosure. Rather, these example embodiments are provided so that this disclosure will satisfy applicable legal requirements. Like reference numerals refer to like elements throughout. Furthermore, as used herein, the term “or” is to be interpreted as a logical operator that results in true whenever one or more of its operands are true. As used herein, operable coupling should be understood to relate to direct or indirect connection that, in either case, enables functional interconnection of components that are operably coupled to each other.
(14) According to some example embodiments, an example torque wrench is described herein that measures a torque applied to a fastener accurately, regardless of where the turning force is applied to the handle of the torque wrench (e.g., the hand position). In this regard, according to some example embodiments, two or more strain gauges may be integrated into the torque wrench that facilitate the ability make accurate torque measurements regardless of the location of the applied force. For example, two strain gauges may be included on a deflection member of the torque wrench that is affixed the drive head. The deflection member may also be affixed to the handle of the torque wrench at two locations, which may be referred to as loading points because the force applied to the handle is transferred to the deflection member at these affixing locations.
(15) The strain gauges may be positioned on the deflection member in relation to the axis of rotation of the drive head (i.e., the drive axis) and the two loading points. According to some example embodiments, the first strain gauge may be positioned on the deflection member rearward of the axis of rotation of the drive head and forward of the first loading point. The second strain gauge may be positioned on the deflection member rearward of the first loading point and forward of the second loading point. According to some example embodiments, the positioning of the strain gauges may be defined based on equivalent distancing ratios. In this regard, for example, a ratio of the distance between the drive axis and the first strain gauge divided by the distance between the

drive axis and the first loading point may be equal to a ratio of the distance between the second loading point and the second strain gauge divided by the distance between the first loading point and the second loading point. According to some example embodiments, positioning the strain gauges in accordance with these ratios, and electrically connecting the strain sensors of the strain gauges as described herein may permit accurate measurement of the torque applied to the fastener, regardless of the position of the force that is applied to the handle. In this regard, each strain gauge may comprise one or more strain sensors.

(16) To further describe these aspects, as well as others, reference is now made to FIG. **1**, which illustrates an example torque wrench **100** according to some example embodiments. For reference, the torque wrench **100** may have a forward end **200** and a rearward end **210**. The torque wrench **100** comprises a drive head **110** and a handle **120**. The drive head **110** may be disposed at the forward end **200** of the torque wrench **100** and, for example, may be a ratcheting drive head comprising a drive tang **114** and a reversing lever **112**. In this regard, the drive head **110** and the drive tang **114** may rotate about an axis of rotation (i.e., the drive axis **230**). The drive head **110**, and more specifically the drive tang **114** may be configured to engage with a tool, such as, for example, a socket or a driver bit, that can engage with, and rotate, a corresponding fastener (e.g., a bolt, nut, screw, or the like). Based on the positioning of the reversing lever **112**, the drive tang **114** may be configured to apply a fastener driving force in a first rotational direction and freely ratchet in a second rotational direction that is opposite the first rotational direction.

(17) The handle **120** may be an elongate member that extends along a longitudinal axis **240** of the torque wrench **100**. The handle **120** may be disposed rearward of the drive head **110** and may be coupled to the drive head **110** via the deflection member **150** (FIG. **2**) at a forward end of the handle **120** via, for example, pins **132** and **134** as further described below. The handle **120** may also include a user interface **350** that may be disposed on the handle **120**. In this regard, the user interface **350** may comprise a display **352** and a keypad **354**. The display **352** may be controlled by processing circuitry (further described below) to provide visual information such as, for example, a torque measurement indicating an amount of torque currently being applied to a fastener by the torque wrench **100**. According to some example embodiments, the display **352** may also be an input device, in the form of, for example, a touch screen display. The keypad **354** may comprise one or more keys or buttons that permit a user to input data for receipt by the processing circuitry. In this manner, for example, the user may input data indicating a torque threshold setting.

(18) Now referring to FIG. **2**, an exploded view of the forward portion of the torque wrench **100** is illustrated, providing a view of the deflection member **150**. In this regard, the deflection member **150** may be an elongate extension that is coupled to the drive head **110** at a forward end of the deflection member **150**. The deflection member **150** may include four surfaces that extend along a length of the deflection member **150**. In this regard, the deflection member **150** may have a back face **241** and a front face (not visible) opposite the back face **241**, each of which are disposed on respective planes that are orthogonal to the drive axis **230**. The deflection member **150** may also include a first side **242** and a second side **243**. The first side **242** may be disposed on an opposite side of the deflection member **150** from the second side **243**. Additionally, the first side **242** and the second side **243** may be positioned such that the first side **242** and the second side **243** are under a strain force when a moment is applied about the drive axis **230**. In this regard, the first side **242** and the second side **243** may be defined on respective planes that are parallel to the drive axis **230**.

(19) As mentioned above, to provide added leverage to the user when using the torque wrench **100**, the deflection member **150** may be coupled to an outer body or handle **120**. In this regard, the outer body may comprise the handle **120**. The handle **120** may be a tubular member with a longitudinally directed opening at the forward end of the handle **120**. The deflection member **150** may be inserted into the opening in the forward end of the handle **120**, and rigidly coupled or affixed to the handle **120** at a first loading point and a second loading point (e.g., points of mechanical coupling). In this regard, the loading points may be locations where the deflection member **150** is coupled or affixed

to the handle **120**. According to some example embodiments, the loading points may be positioned centrally on the deflection member **150** to couple the front face and the back face of the deflection member **150** to the handle **120** at the loading points. According to some example embodiments, the only mechanical coupling between the deflection member **150** and the handle **120** is at the loading points. According to some example embodiments, the loading points may be positioned to be in linear alignment with each other and the drive axis **230**.

(20) According to some example embodiments, to form the loading points, the deflection member **150** may include openings **152** and **154** that may pass through the deflection member **150** from the back face **241** to the front face. The handle **120** may have corresponding openings **124** and **126** on the back face of the handle **120**, as well as corresponding openings on the front face of the handle **120**. As such, when the deflection member **150** is inserted into the longitudinal opening in the forward end of the handle **120**, the openings **152** and **154** may be aligned with the corresponding openings **124** and **126** (as well as the openings on the front face of the handle **120**). With these openings aligned, according to some example embodiments, the pin **132** may be inserted through the respective openings in the handle **120** and the deflection member **150** and secured in place with the ring lock **136** to form the first loading point **252** (FIG. **4**) of the torque wrench **100**. Similarly, according to some example embodiments, the pin **134** may be inserted through the respective openings in the handle **120** and the deflection member **150** and secured in place with the ring lock **138** to form the second loading point **254** (FIG. **4**). According to some example embodiments, while the use of pins **132** and **134** may be one manner to affix the deflection member **150** to the handle **120** to form the first loading point **252** and the second loading point **254**, it is understood that any type of mechanical coupling technique may be used to affix the handle **120** to the deflection member **150** at two points to form the first loading point **252** and the second loading point **254** at their respective positions.

(21) According to some example embodiments, a plurality of strain gauges may be disposed on or integrated with the deflection member **150**. While example embodiments described herein comprise two strain gauges, it is contemplated that more than two strain gauges may be utilized according to some example embodiments. Referring the example embodiments as shown in FIG. **2**, a first strain gauge **161** and a second strain gauge **165** may be disposed on, or integrated with, the deflection member **150**. The first strain gauge **161** may comprise one or more strain sensors, and the second strain gauge **165** may comprise one or more strain sensors. In this example embodiment of the torque wrench **100** shown in FIGS. **2** through **7**, the torque wrench **100** is shown with the first strain gauge **161** comprising two strain sensors and the second strain gauge **165** comprising two strain sensors. However, it is understood and contemplated that a strain gauge, according to some example embodiments, may comprise any number of strain sensors to determine a strain on the deflection member **150** at a position of the strain gauge.

(22) In this regard, the first strain gauge **161** may comprise a first strain sensor **160** and a second strain sensor **162**. Similarly, the second strain gauge **165** may comprise a third strain sensor **164** and a fourth strain sensor **166**. According to some example embodiments, each of the strain sensors may be resistive elements that change an electrical resistance across the strain sensor in proportion with an amount of strain that is applied to the strain sensor. Accordingly, when affixed to a surface (e.g., a surface of the deflection member **150**), an electrical resistance of the strain sensor may change based on the amount of strain being applied to the surface where the strain sensor has been applied. According to some example embodiments, the strain sensors may be formed as conductive traces that are applied or deposited on to the deflection member **150**. The conductive traces may have a serpentine-shape that causes the electrical resistance across the conductive trace to change as a function of the strain applied to the conductive trace. Due to the known relationship between the electrical resistance and the applied strain, a measurement of the strain applied to the strain sensor can be determined based on the electrical resistance.

(23) As such, the first strain gauge **161**, with the first strain sensor **160** and the second strain sensor

162, may be disposed on the deflection member **150** at a position between the drive axis **230** and the first loading point **252** (as defined by the position of the pin **132**). The first strain sensor **160** may be disposed on the first side **242** of the deflection member **150** and the second strain sensor **162** may be disposed on the second side **243** of the deflection member **150**. The first strain sensor **160** and the second strain sensor **162** may be disposed such that the first strain sensor **160** and the second strain sensor **162** are symmetrical about a first strain gauge alignment axis **167** that passes centrally through first strain sensor **160**, the second strain sensor **162**, and the deflection member **150** (from the first side **242** to the second side **243**) and is orthogonal to the drive axis **230**. According to some example embodiments, to protect the first strain sensor **160** and the second strain sensor **162** from interaction with the interior surfaces of the handle **120**, the first strain sensor **160** and the second strain sensor **162** may be disposed in respective recesses on the first side **242** and second side **243** of the deflection member **150**.

(24) Similarly, the second strain gauge **165**, with the third strain sensor **164** and the fourth strain sensor **166**, may be disposed on the deflection member **150** at a position between the first loading point **252** (as defined by the position of the pin **132**), and the second loading point **254** (as defined by the position of the pin **134**). The third strain sensor **164** may be disposed on the first side **242** of the deflection member **150** and the fourth strain sensor **166** may be disposed on the second side **243** of the deflection member **150**. The third strain sensor **164** and the fourth strain sensor **166** may be disposed such that the third strain sensor **164** and the fourth strain sensor **166** are symmetrical about a second strain gauge alignment axis **169** that passes centrally through third strain sensor **164**, the fourth strain sensor **166**, and the deflection member **150** (from the first side **242** to the second side **243**) and is orthogonal to the drive axis **230**. According to some example embodiments, to protect the third strain sensor **164** and the fourth strain sensor **166** from interaction with the interior surfaces of the handle **120**, the third strain sensor **164** and the fourth strain sensor **166** may be disposed in respective recesses on the first side **242** and second side **243** of the deflection member **150**.

(25) Now referring to FIG. **3**, a cross-section view of the torque wrench **100** is shown in a fully assembled configuration. As such, the first strain gauge **161** (comprising the first strain sensor **160** and the second strain sensor **162**) is shown in a position disposed between the drive axis **230** (indicated by a dot due to the orientation of the torque wrench **100** in FIG. **3**) and the first loading point **252**. Additionally, the positioning of the second strain gauge **165** (comprising the third strain sensor **164** and the fourth strain sensor **166**) is shown in a position disposed between the first loading point **252** and the second loading point **254**. An electronics assembly **300** is also shown as being disposed within the handle **120**, with an electrical connection to the first strain gauge **161** and the second strain gauge **165** disposed on the deflection member **150**.

(26) With reference to FIGS. **4** and **5**, an example embodiment of the torque wrench **100** and the deflection member **150** are shown in association with forces resulting from a turning force F applied to handle **120**. Additionally, the distances between various elements are defined to facilitate description of the positioning and relationships between the elements. With respect to the distances between the elements, a defines a distance between the drive axis **230** and the first strain gauge **161** (or more specifically, the first strain gauge alignment axis **167**); b defines a distance between the second loading point **254** and the second strain gauge **165** (or more specifically, the second strain gauge alignment axis **169**); m defines a distance between drive axis **230** and the first loading point **252**; and l defines a distance between the first loading point **252** and the second loading point **254**.

(27) With respect to the applied force F, X defines a distance between the drive axis **230** and a point **250** on the handle **120** where the force F is applied. Force F is applied in a circumstance where the torque wrench **100** is engaged with, for example, a fastener to generate a moment M at the drive axis **230** where the system of forces result in equilibrium (i.e., moment and force equilibrium). Because force F, as shown in FIG. **4**, is applied in an downward direction, translation of the force F through the handle **120** results in component forces of F being applied to the deflection member

150 at the first loading point **252** and the second loading point **254**. In this regard, the component force F**1** applied by the handle **120** at the first loading point **252** and is also directed is directed downward. Further, the component force applied by the handle **120** at the second loading point **254** is defined as F**2** and is directed upward based on the downward orientation of the force F.

(28) Based on the moment equilibrium of the torque wrench **100**, the relationship at the first loading point **252** may be represented by:

$F*(X-m)=F2*l$ (1).

(29) Similarly, due to the force equilibrium of the system, the relationship of the forces may be represented by:

$F2=F+F1$ (2)

Based on (1) and (2) the following relationships can be defined as follows:

(30) $F1=\frac{F*(X-m-l)}{l}$ (3) and: $F2=\frac{F*(X-m)}{l}$. (4)

(31) Referring now to FIG. **5**, the force diagram with respect to the deflection member **150** is shown, and it is noted that forces F**1** and F**2** are oriented in opposite directions relative the forces applied by the handle **120** in FIG. **4**. Accordingly, the moments at the first strain gauge **161** and the second strain gauge **165** may be defined with reference to the first strain gauge alignment axis **167** and the second strain gauge alignment axis **169**. In this regard, the bending moment M**1** at the first strain gauge alignment axis **167** may be defined as:

$M1=F2*(l+m-a)-F1*(m-a)$ (5).

Substituting (3) and (4) into equation (5) yields:

$M1=F*(X-a)$ (6).

(32) Similarly, the bending moment M**2** at the second strain gauge alignment axis **169** may be defined as:

$M2=F2*b$ (7).

Substituting (4) into equation (7) yields:

$M2=F*(X-m)*b/l$ (8).

(33) As such, using on the relationships defined in equations (7) and (8) the difference in the moments can be defined as:

(34) $M1-M2=F*X*(1-\frac{b}{l})+F*(\frac{m*b}{l}-a)$. (9)

Additionally, since the torque applied by the torque wrench **100** may be defined as T=F*X, then element

(35) $(\frac{m*b}{l}-a)$

can be set to be equal to zero, according to some example embodiments, and thus the resulting relationship can be defined as:

(36) $M1-M2=T*(1-\frac{b}{l})$. (10)

(37) Based on equation (10), the difference between the strain applied to the first strain gauge **161** and the strain applied to the second strain gauge **165** is not dependent upon the position of the force F, if, as provided in the following:

(38) $\frac{a}{m}=\frac{b}{l}$. (11)

As such, according to some example embodiments, when this condition, as provided in equation (11), is met by the architecture of the torque wrench **100**, a torque measurement can be determined accurately without regard to the location of the applied force F on the handle **120** rearward of the first loading point **252**, for example, as a function of the bending moments applied to the first strain gauge **161** and the second strain gauge **165**.

(39) In this regard, according to some example embodiments, even in instances where a handle extension (e.g., cheater bar) is used, the torque may be measured accurately. In this regard, with reference to FIG. **6**, the torque wrench **100** is shown with a handle extender **180** coupled to the handle **120** of the torque wrench **100**. The handle extender **180** may be a removable member that be

added to the torque wrench **100** when, for example, added leverage is needed to be applied to a fastener. As such, the handle extender **180** may be configured to be removable from the handle **120** by a user or installed on the handle **120** by the user, when the user wishes to add length to the handle **120**. As such, the handle extender **180** may be configured to increase a handle length of the torque wrench **100** relative to the handle length without the handle extender **180** coupled to the handle **120**. In this regard, the handle length may be defined as a length from the drive axis **230** to a rearward end of the torque wrench **100**, which may be the rearward end **210** of the handle **120** when no handle extender **180** is installed on the torque wrench **100** or the rearward end **250** of the handle extender **180** when the handle extender **180** is installed on the torque wrench **100**. In this regard, the handle extender **180** may be configured to increase a handle length of the torque wrench **100**.

(40) The handle extender **180** may, for example, be an open-ended tube that may receive the rearward end **210** of the handle **120** into the opening of the handle extender **180**. According to some example embodiments, the handle extender **180** may be coupled to the handle **120** via pins **182** and **184** that may pass through the handle extender **180** and the handle **120** to secure the handle extender to the handle **120**. Other means for removably coupling the handle extender **180** to the handle **120**, such as, for example, a detent engagement between the handle **120** and the handle extender **180**, the handle extender **180** may press fit onto the handle **120**, the handle extender **180** may be threaded such that the handle extender **180** screws onto handle **120**, to the like. The handle extender **180** may extend a length of the torque wrench **100** by a distance h and may allow for a force F to be placed further away from the drive axis **230** on the handle extender **300** (i.e., allowing the distance X to extend onto the handle extender **180**). However, because torque measurement can be determined accurately without regard to the location of the applied force F on the handle **120** or the even on the handle extender **180**, rearward of the first loading point **252**, the use of the handle extender **180** and an applied force F on the handle extender **180** does not impact the accuracy of the torque measurements.

(41) As such, according to some example embodiments, the architecture of the torque wrench **100** may be defined in association with the distances, and the relationships between the distances described above. In this regard, a first ratio (a/m) may be defined as a first distance (a) defined between the drive axis **230** and the first strain gauge **161** (e.g., the first strain gauge alignment axis **167**) divided by a second distance (m) defined between the drive axis **230** and the first loading point **252**. The first ratio (a/m) may be equal to a second ratio (b/l) of a third distance (b) defined between the second loading point **254** and the second strain gauge **165** (e.g., the second strain gauge alignment axis **169**) divided by a fourth distance (l) defined between the first loading point **252** and the second loading point **254**.

(42) As described above, the first strain gauge **161** and the second strain gauge **165** may comprise respective strain sensors that may be embodied as resistive elements that change resistance in proportion to the applied strain on the sensors. FIG. **7** illustrates an electrical schematic configuration of the first strain gauge **161** and the second strain gauge **165** (collectively referred to as the strain gauge assembly **163**). In this regard, first strain gauge **161** may comprise the first strain sensor **160** represented as resistance value R**1** and the second strain sensor **162** represented as resistance value R**2**. The second strain gauge **165** may comprise third strain sensor **164** represented as resistance value R**3** and fourth strain sensor **166** represented as resistance value R**4**.

(43) As shown in the strain gauge assembly **163** of FIG. **7**, the first strain gauge **161** may be electrically connected in parallel with the second strain gauge **165**. Further, the first strain sensor **160** may be electrically connected in series with the second strain sensor **162**. Similarly, the third strain sensor **164** may be connected in electrical series with the fourth strain sensor **166**. The first strain gauge **161** may also define a first measurement node **172** between the first strain sensor **160** and the second strain sensor **162**, and the second strain gauge **165** may define a second measurement node **174** between the third strain sensor **164** and the fourth strain sensor **166**.

(44) In operation, a known voltage U.sub.in **176** may be applied across the parallel connected first strain gauge **161** and second strain gauge **165**. The known voltage U.sub.in **176** may be furnished by, for example, a battery or the like. A voltage U.sub.out **178** may be an output that is based on the strain being applied the first strain gauge **161** and the second strain gauge **165**. According to some example embodiments, an output or an output voltage of the strain gauge assembly **163** may be the voltage U.sub.out **178**. Further, an output voltage of the first strain gauge **161** may be a voltage measured between the first measurement node **172** and a ground node of the electronics assembly **300**. An output voltage of the second strain gauge **165** may be a voltage measured between the second measurement node **174** and a ground node of the electronics assembly **300**. As further described below, processing circuitry may be electrically connected across the first measurement node **172** and the second measurement node **174** to measure the voltage U.sub.out **178** from the strain gauge assembly **163** for use in determining a torque measurement indicative of the amount of torque being applied by the torque wrench **100** on a fastener.
(45) Due to electrical architecture of the strain sensors the following relationships may be defined. In this regard, since the first strain sensor **160** is disposed directly opposite the second strain sensor **162** on opposite sides of the deflection member **150**, the resistance value of R**1** may be the same value as R**2**, but with an opposite direction or sign. Similarly, since the third strain sensor **164** is disposed directly opposite the fourth strain sensor **166** on opposite sides of the deflection member **150**, the resistance value of R**3** may be the same value as R**4**, but with an opposite direction or sign. Based on these relationships appropriate substitutions may be made for determining the relationship between the strain on the strain sensors and the voltage U.sub.out **178**. In this regard, U.sub.out may be defined as:
U.sub.out=U.sub.ab=U.sub.a−U.sub.b (12).
In this regard, U.sub.a is the voltage between the first measurement node **172** and ground, U.sub.b is the voltage between the second measurement node **174** and ground, and U.sub.ab is a notation for the difference of U.sub.a and U.sub.b. Substituting the resistance values into (12) yields:
(46) $U_{\text{out}} = (\frac{R+\ R1}{R+\ R1+R-\ R1} - \frac{R+\ R3}{R+\ R3+R-\ R3})U_{in}$. (13)
In this regard, R is the resistance of a strain sensor at no deflection condition, ΔR**1** is the resistance change in strain sensors **160** and **162**, ΔR**3** is the resistance change of strain sensors **164** and **166**, and U.sub.in is the voltage applied across the nodes at **176**.
(47) Further simplifying (13) for the resistance values yields:
(48) $U_{\text{out}} = \frac{1}{2}(\frac{R1}{R} - \frac{R3}{R})U_{in} = \frac{1}{2}K(\ 1 - \ 3)U_{in}$. (14)
In this regard, ε1 is the strain of the deflection member **150** at the location of first strain gauge **161**, and ε3 is the strain of the deflection member **150** at the location of second strain gauge **165**. Additionally, K is the sensitivity of the strain gauges (i.e., first strain gauge **161** and second strain gauge **165**).
(49) Again, further substitutions yields, in terms of bending moments:
(50) $U_{\text{out}} = \frac{1}{2}K(\frac{M1}{W*E} - \frac{M2}{W*E})U_{in} = \frac{K}{2*W*E}(M1 - M2)U_{in}$. (15)
In this regard, M**1** is the bending moment at the location of first strain gauge **161**, and M**2** is the bending moment at the location of second strain gauge **165**. Further, W is the section modulus in bending of the deflection member **150**, and E is the elasticity modulus of the defection member **150**.
(51) Finally, substituting from (10) yields:
(52) $U_{\text{out}} = \frac{K*T}{2*W*E}(1 - \frac{b}{l})U_{in}$. (16)
(53) As such, the relationship between the voltage U.sub.out **178** and the torque T can be defined and used for determining measurements of applied torque by, for example, the processing circuitry of the torque wrench **100**. Accordingly, the torque T may be determined as a function of the voltage measured across the first measurement node **172** and the second measurement node **174** (i.e.,

voltage U.sub.out **178**). As such, the torque T may have a defined relationship with the measured voltage.
(54) Now referring to FIG. **8**, an electronics assembly **300** of the torque wrench **100** is shown that may be configured to perform various functionalities associated with the operation of the torque wrench **100**. In this regard, the electronics assembly **300**, and more specifically, the processing circuitry **310**, may be configured to measure an output voltage of the strain gauge assembly **163** (i.e., voltage U.sub.out **178**) and convert the voltage measurement to a torque measurement for the torque wrench **100**.
(55) In this regard, FIG. **8** illustrates a block diagram of the electronics assembly **300**, according to some example embodiments. The electronics assembly **300** comprises processing circuitry **310**, which may comprise a processor **320**, a memory **330**, and a user interface **350**. Further, the electronics assembly **300** is not limited and may include additional components not shown in FIG. **8** and the processing circuitry **310** may be operably coupled to other components of the torque wrench **100** that are not shown in FIG. **8**.
(56) According to some example embodiments, processing circuitry **310** may be in operative communication with or embody, the memory **330**, the processor **320**, and the user interface **350**. Through configuration and operation of the memory **330**, the processor **320**, and the user interface **350**, the processing circuitry **310** may be configurable to perform various operations as described herein, including the operations and functionalities described with respect to the torque wrench **100** and the strain gauge assembly **163**. In this regard, the processing circuitry **310** may be configured to perform computational processing, memory management, user interface control and monitoring, and the like, according to an example embodiment. In some embodiments, the processing circuitry **310** may be embodied as a chip or chip set. In other words, the processing circuitry **310** may comprise one or more physical packages (e.g., chips) including materials, components, or wires on a structural assembly (e.g., a baseboard or printed circuit board). The processing circuitry **310** may be configured to receive inputs (e.g., via peripheral components), perform actions based on the inputs, and generate outputs (e.g., for provision to peripheral components). In an example embodiment, the processing circuitry **310** may include one or more instances of a processor **320**, associated circuitry, and memory **330**. As such, the processing circuitry **310** may be embodied as a circuit chip (e.g., an integrated circuit chip, such as a field programmable gate array (FPGA)) configured (e.g., with hardware, software or a combination of hardware and software) to perform operations described herein.
(57) In an example embodiment, the memory **330** may include one or more non-transitory memory devices such as, for example, volatile or non-volatile memory that may be either fixed or removable. The memory **330** may be configured to store information, data, applications, instructions or the like for enabling, for example, the functionalities described with respect to the torque wrench **100**. The memory **330** may operate to buffer instructions and data during operation of the processing circuitry **310** to support higher-level functionalities, and may also be configured to store instructions for execution by the processing circuitry **310**. The memory **330** may also store various information including torque measurements, torque threshold settings, or the like. According to some example embodiments, various data stored in the memory **330** may be generated based on other data and stored in the memory **330** such as, for example, voltage measurements.
(58) As mentioned above, the processing circuitry **310** may be embodied in a number of different ways. For example, the processing circuitry **310** may be embodied as various processing means such as one or more processors **320** that may be in the form of a microprocessor or other processing element, a coprocessor, a controller or various other computing or processing devices including integrated circuits such as, for example, an ASIC (application specific integrated circuit), an FPGA, or the like. In an example embodiment, the processing circuitry **310** may be configured to execute instructions stored in the memory **330** or otherwise accessible to the processing circuitry **310**. As

such, whether configured by hardware or by a combination of hardware and software, the processing circuitry **310** may represent an entity (e.g., physically embodied in circuitry—in the form of processing circuitry **310**) capable of performing operations according to example embodiments, while configured accordingly. Thus, for example, when the processing circuitry **310** is embodied as an ASIC, FPGA, or the like, the processing circuitry **310** may be specifically configured hardware for conducting the operations described herein. Alternatively, as another example, when the processing circuitry **310** is embodied as an executor of software instructions, the instructions may specifically configure the processing circuitry **310** to perform the operations described herein.

(59) The user interface **350** may be controlled by the processing circuitry **310** to interact with peripheral components or devices of the torque wrench **100** that can receive inputs from a user or provide outputs to a user. In this regard, via the user interface **350**, the processing circuitry **310** may be configured to receive inputs from an input device which may be, for example, a touch screen display (e.g. display **352**), a keypad **354**, a microphone, camera or the like. The user interface **350** may also be configured to provide control and outputs to peripheral devices such as, for example, the display **352**, an audible/haptic feedback device **356**, or the like. The user interface **350** may also produce outputs, for example, as visual outputs on a display, audio outputs via a speaker, or the like. The audible/haptic feedback device **356** may be a sounder, speaker, vibrator, or the like that can provide sensory feedback to a user. In this regard, according to some example embodiments, the audible/haptic feedback device **356** may be configured to alert the user by providing an audible tone or a vibration when a measured torque of the torque wrench **100** is equal to or exceed a torque threshold setting, which may be input by the use via the keypad **354** and the display **352**.

(60) As such, according to some example embodiments, the processing circuitry **310** may be operably coupled to the strain gauge assembly **163** to measure an output voltage of the strain gauge assembly **163**. More specifically, according to some example embodiments, the processing circuitry **310** may be electrically connected to outputs of the first strain gauge **161** and the second strain gauge **165** in the form of the first measurement node **172** and the second measurement node **174**. In this regard, the processing circuitry **310** may be configured to measure a voltage (e.g., the voltage U.sub.out **178**) via the electrical connections to the first measurement node **172** and the second measurement node **174**.

(61) Additionally, according to some example embodiments, the processing circuitry **310** may be configured to generate a torque measurement based on the voltage measured from the output of the strain gauge assembly **163** or measured across the first measurement node **172** and the second measurement node **174**. The measured voltage may be used in association with the relationships and equations described above to convert or calculate a torque measurement, for example, as a function of the measured voltage. According to some example embodiments, the processing circuitry **310** may be configured to, via the user interface **350**, control the display **352** to output the torque measurement on a display **352**.

(62) Further, according to some example embodiments, the processing circuitry **310** may be configured to receive a torque threshold setting from a user, for example, via the keypad **354**. The processing circuitry **310** may be configured to store the torque threshold setting in, for example, the memory **330**. Further, the processing circuitry **310** may be configured to periodically determine a current torque measurement being applied to a fastener by the torque wrench **100** and compare the current torque measurement to the torque threshold setting, for example, to determine when the torque applied by the torque wrench **100** on the fastener is equal to or exceeds the torque threshold setting. According to some example embodiments, the torque threshold setting may be stored in the memory **330** in a form that can be directly compared to the voltage measured from the output of the strain gauge assembly **163**, which may avoid having to convert the voltage measurement each time a comparison is performed. As such, the measured voltage or a conversion of the measured voltage may be compared to the torque threshold setting. According to some example embodiments, in

response to the measured voltage or a conversion of the measured voltage being equal to or exceeding the torque threshold setting, the processing circuitry **310** may be configured to output a feedback alert (e.g., a sound, vibration, visual indicator, or the like), for example, via the audible/haptic feedback device **356** or the display **352** of the user interface **350**.

(63) Now referring to FIG. **9**, a flowchart of an example method for measuring a torque applied by a drive head of a torque wrench to a fastener is provided. In this regard, the torque wrench (e.g., torque wrench **100**) may comprise a deflection member coupled to the drive head. The example method may comprise, at **400**, measuring, by processing circuitry, a voltage between an output of a first strain gauge (e.g., first measurement node **172**) and an output of a second strain gauge (e.g., second measurement node **174**). The measured voltage may be based on a torque being applied to the fastener. The first strain gauge may be coupled to the deflection member between the drive head and a first loading point. The second strain gauge may be coupled to the deflection member between the first loading point and a second loading point. The first loading point and the second loading point may be points of mechanical coupling between the deflection member and a handle of the torque wrench. According to some example embodiments, the example method may also include converting the measured voltage into a torque measurement at **410**.

(64) According to some example embodiments, the example method may further comprise, at **420**, outputting the torque measurement on a display. Additionally or alternatively, the example method may include, at **430**, comparing the measured voltage or a conversion of the measured voltage to a torque threshold setting, and, at **440**, controlling a user interface to output a feedback alert to a user when the measured voltage or a conversion of the measured voltage is equal to the torque threshold setting.

(65) Based on the forgoing, according to some example embodiments, an improved torque wrench **100** is provided that can improve the accuracy of torque measurements when a force is applied at different locations on the torque wrench **100**. In this regard, FIG. **10** illustrates a graph **500** showing the variation in measurements that can occur using a conventional torque wrench that does not incorporate aspects of the example embodiments described herein. As can be seen in FIG. **10**, as the applied force moves closer to the center of the driver/the drive axis, the accuracy of the torque measurements begin to degrade rapidly.

(66) FIG. **11** illustrates the performance of a torque wrench, such as torque wrench **100**, that has incorporated aspects of the example embodiments described herein. In this regard, FIG. **11** shows a graph **600** of torque measurements taken with different applied torques and at different positions on the torque wrench. As can be seen, unlike the graph **500**, the torque measurements remain consistently accurate, even as the force position moves closer to the center of the driver/the drive axis.

(67) Having described various aspects of example embodiments, some additional example embodiments will now be described. According to some example embodiments, an example torque wrench is provided. The torque wrench may comprise a drive head configured to engage with a tool for rotating a fastener. The drive head may have a drive axis about which the drive head rotates when rotating the fastener. Further, the torque wrench may comprise a deflection member coupled to the drive head, and an outer body coupled to the deflection member at a first loading point and a second loading point. The torque wrench may also comprise a first strain gauge coupled to the deflection member between the drive head and the first loading point, and a second strain gauge coupled to the deflection member between the first loading point and the second loading point.

(68) Additionally, according to some example embodiments, a first ratio of a first distance between the drive axis and the first strain gauge divided by a second distance between the drive axis and the first loading point is equal to a second ratio of a third distance between the second loading point and the second strain gauge divided by a fourth distance between the first loading point and the second loading point. Additionally or alternatively, according to some example embodiments, the first strain gauge may comprise a first strain sensor and a second strain sensor. The second strain

gauge may comprise a third strain sensor and a fourth strain sensor. The first strain sensor may be disposed on a first side of the deflection member and the second strain sensor may be disposed on a second side of the deflection member. The first side of the deflection member may be opposite the second side of the deflection member. Further, the third strain sensor may be disposed on the first side of the deflection member and the fourth strain sensor may be disposed on the second side of the deflection member.

(69) Additionally or alternatively, according to some example embodiments, the first strain sensor and second strain sensor may be symmetrical about a first strain gauge alignment axis, and the third strain sensor and the fourth strain sensor are symmetrical about a second strain gauge alignment axis. Additionally or alternatively, an electrical resistance of the first, second, third, or fourth strain sensor may vary in proportion to an amount of strain applied to the deflection member at a location where the respective strain sensor is coupled to the deflection member. Additionally or alternatively, the first strain sensor is electrically connected in series with the second strain sensor, and the third strain sensor is electrically connected in series with the fourth strain sensor. Additionally or alternatively, the first strain gauge may define a first measurement node disposed electrically between the first strain sensor and the second strain sensor, and the second strain gauge may define a second measurement node disposed electrically between the third strain sensor and the fourth strain sensor. Additionally or alternatively, the torque wrench may comprise processing circuitry operably coupled to the first measurement node and the second measurement node. The processing circuitry may be configured to generate a torque measurement based on a voltage measured between the first measurement node and the second measurement node. Additionally or alternatively, the first strain gauge and the second strain gauge may be electrically connected in parallel. Additionally or alternatively, the outer body may comprise an elongate handle. The elongate handle may be coupled to a handle extender. The handle extender may be configured to increase a handle length of the torque wrench. Additionally or alternatively, the outer body may be coupled to the deflection member at the first loading point by a first pin that passes through a first opening in the outer body and a first opening in the deflection member, and the outer body may be coupled to the deflection member at the second loading point by a second pin that passes through a second opening in the outer body and a second opening in the deflection member.

(70) According to some example embodiments, another example torque wrench is provided. The torque wrench may comprise a drive head configured to engage with a tool for rotating a fastener. The drive head may have a drive axis about which the drive head rotates when rotating the fastener. The torque wrench may also comprise a deflection member coupled to the drive head and a handle coupled to the deflection member at a first loading point and a second loading point. The torque wrench may further comprise a first strain gauge coupled to the deflection member between the drive head and the first loading point, and a second strain gauge coupled to the deflection member between the first loading point and the second loading point. Additionally, the torque wrench may comprise processing circuitry electrically coupled to the first strain gauge and the second strain gauge and configured to measure a voltage between an output of the first strain gauge and an output of the second strain gauge. The voltage may be based on and have a quantitative relationship to a torque being applied to the fastener.

(71) Additionally, according to some example embodiments, the processing circuitry may be configured to convert the measured voltage into a torque measurement, and output the torque measurement on a display. Additionally or alternatively, according to some example embodiments, the processing circuitry may be further configured to compare the measured voltage or a conversion of the measured voltage to a torque threshold setting, and control a user interface to output a feedback alert when the measured voltage or a conversion of the measured voltage is equal to the torque threshold setting. Additionally or alternatively, according to some example embodiments, a first ratio of a first distance between the drive axis and the first strain gauge divided by a second distance between the drive axis and the first loading point is equal to a second

ratio of a third distance between the second loading point and the second strain gauge divided by a fourth distance between the first loading point and the second loading point. Additionally or alternatively, according to some example embodiments, the first strain gauge and the second strain gauge are electrically connected in parallel. Additionally or alternatively, the first strain gauge and the second strain gauge may comprise resistive elements having an electrical resistance that varies in proportion to an amount of strain applied to the resistive elements. Additionally or alternatively, according to some example embodiments, the handle may comprise a tube and the deflection member may be disposed within the tube. Additionally or alternatively, according to some example embodiments, the handle may be coupled to a handle extender. The handle extender may be configured to increase a handle length of the torque wrench.

(72) According to some example embodiments, an example method for measuring a torque applied by a drive head of a torque wrench to a fastener is also provided. The torque wrench may comprise a deflection member coupled to the drive head. The method may comprise measuring, by processing circuitry, a voltage between an output of a first strain gauge and an output of a second strain gauge. In this regard, the voltage may based on a torque being applied to the fastener. The first strain gauge may be coupled to the deflection member between the drive head and a first loading point, and the second strain gauge may be coupled to the deflection member between the first loading point and a second loading point. The first loading point and the second loading point may be points of mechanical coupling between the deflection member and a handle of the torque wrench. The method may also comprise converting the measured voltage into a torque measurement.

(73) Additionally, according to some example embodiments, the method may further comprise outputting the torque measurement on a display, comparing the measured voltage or a conversion of the measured voltage to a torque threshold setting, and controlling a user interface to output a feedback alert to a user when the measured voltage or a conversion of the measured voltage is equal to the torque threshold setting. Additionally or alternatively, according to some example embodiments, a first ratio of a first distance between a drive axis of the drive head and the first strain gauge divided by a second distance between the drive axis and the first loading point is equal to a second ratio of a third distance between the second loading point and the second strain gauge divided by a fourth distance between the first loading point and the second loading point.

(74) According to some example embodiments, a torque wrench may comprise a drive head configured to engage with a tool for rotating a fastener. The drive head may have a drive axis about which the drive head rotates when rotating the fastener. The torque wrench may also comprise a deflection member coupled to the drive head, a handle coupled to the deflection member, and a strain gauge assembly coupled to the deflection member. The strain gauge assembly may be configured to measure strain on the deflection member as an indication of a torque being applied to the fastener by the torque wrench. The torque wrench may further comprise a handle extender coupled to the handle. The handle extender may be configured to be removable from the handle by a user or installed on the handle by the user. The handle extender may be configured to increase a handle length of the torque wrench relative to the handle length without the handle extender coupled to the handle. Additionally, the handle may be coupled to the deflection member at a first loading point and a second loading point. Further, the strain gauge assembly may comprise a first strain gauge coupled to the deflection member between the drive head and the first loading point, and a second strain gauge coupled to the deflection member between the first loading point and the second loading point.

(75) Many modifications and other embodiments of the inventions set forth herein will come to mind to one skilled in the art to which these inventions pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the inventions are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended

claims. Moreover, although the foregoing descriptions and the associated drawings describe exemplary embodiments in the context of certain exemplary combinations of elements and/or functions, it should be appreciated that different combinations of elements and/or functions may be provided by alternative embodiments without departing from the scope of the appended claims. In this regard, for example, different combinations of elements and/or functions than those explicitly described above are also contemplated as may be set forth in some of the appended claims. In cases where advantages, benefits or solutions to problems are described herein, it should be appreciated that such advantages, benefits and/or solutions may be applicable to some example embodiments, but not necessarily all example embodiments. Thus, any advantages, benefits or solutions described herein should not be thought of as being critical, required or essential to all embodiments or to that which is claimed herein. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A torque wrench comprising: a drive head configured to engage with a tool for rotating a fastener, the drive head having a drive axis about which the drive head rotates when rotating the fastener; a deflection member coupled to the drive head; an outer body coupled to the deflection member at a first loading point and a second loading point; a first strain gauge coupled to the deflection member between the drive head and the first loading point; and a second strain gauge coupled to the deflection member between the first loading point and the second loading point.
2. The torque wrench of claim 1, wherein a first ratio of a first distance between the drive axis and the first strain gauge divided by a second distance between the drive axis and the first loading point is equal to a second ratio of a third distance between the second loading point and the second strain gauge divided by a fourth distance between the first loading point and the second loading point.
3. The torque wrench of claim 1, wherein the first strain gauge comprises at least a first strain sensor and a second strain sensor; wherein the second strain gauge comprises a third strain sensor and a fourth strain sensor; wherein the first strain sensor is disposed on a first side of the deflection member and the second strain sensor is disposed on a second side of the deflection member, the first side of the deflection member being opposite the second side of the deflection member; and wherein the third strain sensor is disposed on the first side of the deflection member and the fourth strain sensor is disposed on the second side of the deflection member.
4. The torque wrench of claim 3, wherein the first strain sensor and second strain sensor are symmetrical about a first strain gauge alignment axis; and wherein the third strain sensor and the fourth strain sensor are symmetrical about a second strain gauge alignment axis.
5. The torque wrench of claim 3, wherein an electrical resistance of the first strain sensor varies in proportion to an amount of strain applied to the deflection member at a location where the first strain sensor is coupled to the deflection member.
6. The torque wrench of claim 3, wherein the first strain sensor is electrically connected in series with the second strain sensor; and wherein the third strain sensor is electrically connected in series with the fourth strain sensor.
7. The torque wrench of claim 6, wherein the first strain gauge defines a first measurement node disposed electrically between the first strain sensor and the second strain sensor; and wherein the second strain gauge defines a second measurement node disposed electrically between the third strain sensor and the fourth strain sensor.
8. The torque wrench of claim 7 further comprising processing circuitry operably coupled to the first measurement node and the second measurement node; wherein the processing circuitry is configured to generate a torque measurement based on a voltage measured between the first measurement node and the second measurement node.
9. The torque wrench of claim 1, wherein the first strain gauge and the second strain gauge are

electrically connected in parallel.
10. The torque wrench of claim 1, wherein the outer body comprises an elongate handle coupled to a handle extender, the handle extender being configured to increase a handle length of the torque wrench.
11. The torque wrench of claim 1, wherein the outer body is coupled to the deflection member at the first loading point by a first pin that passes through a first opening in the outer body and a first opening in the deflection member; and wherein the outer body is coupled to the deflection member at the second loading point by a second pin that passes through a second opening in the outer body and a second opening in the deflection member.
12. A torque wrench comprising: a drive head configured to engage with a tool for rotating a fastener, the drive head having a drive axis about which the drive head rotates when rotating the fastener; a deflection member coupled to the drive head; a handle coupled to the deflection member at a first loading point and a second loading point; a first strain gauge coupled to the deflection member between the drive head and the first loading point; a second strain gauge coupled to the deflection member between the first loading point and the second loading point; processing circuitry electrically coupled to the first strain gauge and the second strain gauge and configured to measure a voltage between an output of the first strain gauge and an output of the second strain gauge, the measured voltage based on a torque being applied to the fastener.
13. The torque wrench of claim 12, wherein the processing circuitry is configured to convert the measured voltage into a torque measurement, and output the torque measurement on a display.
14. The torque wrench of claim 12, wherein the processing circuitry is further configured to: compare the measured voltage or a conversion of the measured voltage to a torque threshold setting; and control a user interface to output a feedback alert when the measured voltage or a conversion of the measured voltage is equal to the torque threshold setting.
15. The torque wrench of claim 12, wherein a first ratio of a first distance between the drive axis and the first strain gauge divided by a second distance between the drive axis and the first loading point is equal to a second ratio of a third distance between the second loading point and the second strain gauge divided by a fourth distance between the first loading point and the second loading point.
16. The torque wrench of claim 12, wherein the first strain gauge and the second strain gauge are electrically connected in parallel.
17. The torque wrench of claim 12, wherein the first strain gauge and the second strain gauge comprise resistive elements having an electrical resistance that varies in proportion to an amount of strain applied to the resistive elements.
18. The toque wrench of claim 12, wherein the handle is coupled to a handle extender, the handle extender being configured to increase a handle length of the torque wrench.
19. A torque wrench comprising: a drive head configured to engage with a tool for rotating a fastener, the drive head having a drive axis about which the drive head rotates when rotating the fastener; a deflection member coupled to the drive head; a handle coupled to the deflection member; a strain gauge assembly coupled to the deflection member, the strain gauge assembly being configured to measure strain on the deflection member as an indication of a torque being applied to the fastener by the torque wrench; and a handle extender coupled to the handle, the handle extender configured to be removable from the handle by a user and installed on the handle by the user, the handle extender being configured to increase a handle length of the torque wrench relative to the handle length without the handle extender coupled to the handle.
20. A torque wrench comprising: a drive head configured to engage with a tool for rotating a fastener, the drive head having a drive axis about which the drive head rotates when rotating the fastener; a deflection member coupled to the drive head; a handle coupled to the deflection member; a strain gauge assembly coupled to the deflection member, the strain gauge assembly being configured to measure strain on the deflection member as an indication of a torque being

applied to the fastener by the torque wrench; and a handle extender coupled to the handle, the handle extender configured to be removable from the handle by a user or installed on the handle by the user, the handle extender being configured to increase a handle length of the torque wrench relative to the handle length without the handle extender coupled to the handle; wherein the handle is coupled to the deflection member at a first loading point and a second loading point; and wherein the strain gauge assembly comprises a first strain gauge coupled to the deflection member between the drive head and the first loading point, and a second strain gauge coupled to the deflection member between the first loading point and the second loading point.

21. A method for measuring a torque applied by a drive head of a torque wrench too a fastener, the torque wrench comprising a deflection member coupled to the drive head, the method comprising: measuring, by processing circuitry, a voltage between an output of a first strain gauge and an output of a second strain gauge, the voltage being based on a torque being applied to the fastener, the first strain gauge being coupled to the deflection member between the drive head and a first loading point, the second strain gauge being coupled to the deflection member between the first loading point and a second loading point, the first loading point and the second loading point being points of mechanical coupling between the deflection member and a handle of the torque wrench; and converting the measured voltage into a torque measurement.